

Team Note of 팀명고민하기귀찮음

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1 Data Structure for RMQ

1.1 Disjoint Sparse Table

Time Complexity: $\mathcal{O}(N \log N) - \mathcal{O}(1)$, even with non-idempotent operation

```
struct dsparse {
    using T = array<ll, 501>;
    using V = pii;
    static constexpr T E = {};
    static constexpr V I = {};
    int n, level;
    vector<vector<T>> table;
    T merge(T x, V v) {
        for (int i = 500 - v.first; i ≥ 0; --i)
            x[i + v.first] = max(x[i + v.first], x[i] + v.second);
        return x;
    }
    void make(vector<V> a) {
        level = bit_width(a.size()) - 1;
        level = max(level, 1);
        n = 1 << level;
        table.resize(level, vector<T>(a.size(), E));
        function<void(int, int, int)> go = [&](int t, int l, int r) {
            if (l ≥ r)
                return;
            const int m = (l + r) / 2;
            for (int i = m + 1; i ≤ min(r, int(a.size()) - 1); ++i)
                table[t][i] = merge(i > m + 1 ? table[t][i - 1] : E,
                                     a[i]);
            for (int i = min(m, int(a.size()) - 1); i ≥ l; --i)
                table[t][i] = merge(i < m ? table[t][i + 1] : E, a[i]);
            go(t - 1, l, m);
            go(t - 1, m + 1, r);
        };
    }
};
```

```
};
go(level - 1, 0, n - 1);
}
pair<T, T> query(int l, int r) {
    if (l > r) return {E, E};
    if (l == r) return {table[0][l], E};
    const int t = 31 - __builtin_clz(l ^ r);
    return {table[t][l], table[t][r]};
}
};
```

1.2 Persistence Segment Tree

Time Complexity: $\mathcal{O}(\log^2 N)$

```
struct pst {
    struct node {
        dat d = ID;
        array<int, 2> go{};
    };
    int n;
    vector<node> tree;
    vector<int> roots;
    pst(int sz) {
        n = int(ceil(log2(sz)));
        roots.push_back(1);
        tree.resize(1 << (n + 1));
        for (int i = 1; i < (1 << n); ++i) {
            tree[i].go[0] = i * 2;
            tree[i].go[1] = i * 2 + 1;
        }
    }
    int insert(int x, int prev, const dat &value) {
        int curr = tree.size();
        tree.emplace_back();
        roots.push_back(curr);
        vector<int> st;
        for (int i = n - 1; i ≥ 0; --i) {
            st.push_back(curr);
            const int next = x >> i & 1;
            tree[curr].go[next] = tree.size();
        }
    }
};
```

```

    tree.emplace_back();
    tree[curr].go[!next] = tree[prev].go[!next];
    curr = tree[curr].go[next];
    prev = tree[prev].go[next];
}
tree[curr].d = value;
while (!st.empty()) {
    const int x = st.back();
    st.pop_back();
    tree[x].d = tree[tree[x].go[0]].d + tree[tree[x].go[1]].d;
}
return roots.back();
}
dat query(int t, int l, int r) {
    function<dat(int, int, int)> q = [&](int x, int s, int e) ->
    dat {
        if (r < s || e < l)
            return ID;
        if (l ≤ s && e ≤ r)
            return tree[x].d;
        const int m = (s + e) / 2;
        return q(tree[x].go[0], s, m) + q(tree[x].go[1], m + 1, e);
    };
    return q(t, 0, (1 << n) - 1);
}
int walk(int t, int l, int k) {
    function<int(int, int, int)> q = [&](int x, int s, int e) ->
    int {
        if (e < l || tree[x].d.v ≥ k)
            return -1;
        if (s == e)
            return s;
        const int m = (s + e) / 2;
        const int res = q(tree[x].go[0], s, m);
        if (res != -1)
            return res;
        return q(tree[x].go[1], m + 1, e);
    };
    const int ret = q(t, 0, (1 << n) - 1);
    return ret == -1 ? (1 << n) - 1 : ret - 1;
}

```

```

    }
};

```

1.3 Segment Tree Beats

Usage: Note the potential function

Time Complexity: $\mathcal{O}(\log^2 N)$

```

void off(int x, int s, int e, int l, int r, int v) {
    push(x, s, e);
    if (e < l || r < s || (tree[x].o & v) == 0)
        return;
    if (l ≤ s && e ≤ r && !(v & (tree[x].a ^ tree[x].o))) {
        tree[x].l -= v & tree[x].o;
        push(x, s, e);
    } else {
        const int m = (s+e) / 2;
        off(x*2, s, m, l, r, v);
        off(x*2+1, m+1, e, l, r, v);
        tree[x] = tree[x*2] + tree[x*2+1];
    }
}
void on(int x, int s, int e, int l, int r, int v) {
    push(x, s, e);
    if (e < l || r < s || (tree[x].a & v) == v)
        return;
    if (l ≤ s && e ≤ r && !(v & (tree[x].a ^ tree[x].o))) {
        tree[x].l += v & ~tree[x].o;
        push(x, s, e);
    } else {
        const int m = (s+e) / 2;
        on(x*2, s, m, l, r, v);
        on(x*2+1, m+1, e, l, r, v);
        tree[x] = tree[x*2] + tree[x*2+1];
    }
}
int sum(int x, int s, int e, int l, int r) {
    push(x, s, e);
    if (e < l || r < s)
        return 0;
    if (l ≤ s && e ≤ r)

```

```

    return tree[x].x;
    const int m = (s+e) / 2;
    return max(sum(x*2, s, m, l, r), sum(x*2+1, m+1, e, l, r));
}

```

1.4 Fenwick RMQ

Time Complexity: Fast $\mathcal{O}(\log N)$

```

struct fenwick {
    static constexpr pii INF = {1e9 + 7, -(1e9 + 7)};
    vector<pii> tree1, tree2;
    const vector<int> &arr;
    static pii op(pii l, pii r) {
        return {min(l.first, r.first), max(l.second, r.second)};
    }
    fenwick(const vector<int> &a) : arr(a) {
        const int n = a.size();
        tree1.resize(n + 1, INF);
        tree2.resize(n + 1, INF);
        for (int i = 0; i < n; ++i)
            update(i, a[i]);
    }
    void update(int x, int v) {
        for (int i = x + 1; i < tree1.size(); i += i & -i)
            tree1[i] = op(tree1[i], {v, v});
        for (int i = x + 1; i > 0; i -= i & -i)
            tree2[i] = op(tree2[i], {v, v});
    }
    pii query(int l, int r) {
        pii ret = INF;
        l++, r++;
        int i;
        for (i = r; i - (i & -i) >= l; i -= i & -i)
            ret = op(tree1[i], ret);
        for (i = l; i + (i & -i) <= r; i += i & -i)
            ret = op(tree2[i], ret);
        ret = op({arr[i - 1], arr[i - 1]}, ret);
        return ret;
    }
};

```

1.5 Link/Cut Tree

```

struct Node {
    Node *l, *r, *p;
    bool flip;
    int sz;
    T now, sum, lz;
    Node() {
        l = r = p = nullptr;
        sz = 1;
        flip = false;
        now = sum = lz = 0;
    }
    bool IsLeft() const { return p && this == p->l; }
    bool IsRoot() const { return !p || (this != p->l && this != p->r); }
    friend int GetSize(const Node *x) { return x ? x->sz : 0; }
    friend T GetSum(const Node *x) { return x ? x->sum : 0; }
    void Rotate() {
        p->Push();
        Push();
        if (IsLeft())
            r && (r->p = p), p->l = r, r = p;
        else
            l && (l->p = p), p->r = l, l = p;
        if (!p->IsRoot())
            (p->IsLeft() ? p->p->l : p->p->r) = this;
        auto t = p;
        p = t->p;
        t->p = this;
        t->Update();
        Update();
    }
    void Update() {
        sz = 1 + GetSize(l) + GetSize(r);
        sum = now + GetSum(l) + GetSum(r);
    }
    void Update(const T &val) {
        now = val;
        Update();
    }
}

```

```

}
void Push() {
    Update(now + lz);
    if (flip)
        swap(l, r);
    for (auto c : {l, r})
        if (c)
            c->flip ^= flip, c->lz += lz;
    lz = 0;
    flip = false;
}
};
Node *rt;
Node *Splay(Node *x, Node *g = nullptr) {
    for (g || (rt = x); x->p != g; x->Rotate()) {
        if (!x->p->IsRoot())
            x->p->p->Push();
        x->p->Push();
        x->Push();
        if (x->p->p != g)
            (x->IsLeft() ^ x->p->IsLeft() ? x : x->p)->Rotate();
    }
    x->Push();
    return x;
}
Node *Kth(int k) {
    for (auto x = rt;; x = x->r) {
        for (; x->Push(), x->l && x->l->sz > k; x = x->l)
            ;
        if (x->l)
            k -= x->l->sz;
        if (!k--)
            return Splay(x);
    }
}
Node *Gather(int s, int e) {
    auto t = Kth(e + 1);
    return Splay(t, Kth(s - 1))->l;
}
Node *Flip(int s, int e) {

```

```

    auto x = Gather(s, e);
    x->flip ^= 1;
    return x;
}
Node *Shift(int s, int e, int k) {
    if (k ≥ 0) { // shift to right
        k %= e - s + 1;
        if (k)
            Flip(s, e), Flip(s, s + k - 1), Flip(s + k, e);
    } else { // shift to left
        k = -k;
        k %= e - s + 1;
        if (k)
            Flip(s, e), Flip(s, e - k), Flip(e - k + 1, e);
    }
    return Gather(s, e);
}
int Idx(Node *x) { return x->l->sz; }
////////// Link Cut Tree Start //////////
Node *Splay(Node *x) {
    for (; !x->IsRoot(); x->Rotate()) {
        if (!x->p->IsRoot())
            x->p->p->Push();
        x->p->Push();
        x->Push();
        if (!x->p->IsRoot())
            (x->IsLeft() ^ x->p->IsLeft() ? x : x->p)->Rotate();
    }
    x->Push();
    return x;
}
void Access(Node *x) {
    Splay(x);
    x->r = nullptr;
    x->Update();
    for (auto y = x; x->p; Splay(x))
        y = x->p, Splay(y), y->r = x, y->Update();
}
int GetDepth(Node *x) {
    Access(x);

```

```

    x->Push();
    return GetSize(x->l);
}
Node *GetRoot(Node *x) {
    Access(x);
    for (x->Push(); x->l; x->Push())
        x = x->l;
    return Splay(x);
}
Node *GetPar(Node *x) {
    Access(x);
    x->Push();
    if (!x->l)
        return nullptr;
    x = x->l;
    for (x->Push(); x->r; x->Push())
        x = x->r;
    return Splay(x);
}
void Link(Node *p, Node *c) {
    Access(c);
    Access(p);
    c->l = p;
    p->p = c;
    c->Update();
}
void Cut(Node *c) {
    Access(c);
    c->l->p = nullptr;
    c->l = nullptr;
    c->Update();
}
Node *GetLCA(Node *x, Node *y) {
    Access(x);
    Access(y);
    Splay(x);
    return x->p ? x->p : x;
}
Node *Ancestor(Node *x, int k) {
    k = GetDepth(x) - k;

```

```

    assert(k ≥ 0);
    for (;;) x->Push() {
        int s = GetSize(x->l);
        if (s == k)
            return Access(x), x;
        if (s < k)
            k -= s + 1, x = x->r;
        else
            x = x->l;
    }
}
void MakeRoot(Node *x) {
    Access(x);
    Splay(x);
    x->flip ^= 1;
    x->Push();
}
bool IsConnect(Node *x, Node *y) { return GetRoot(x) ==
GetRoot(y); }
void PathUpdate(Node *x, Node *y, T val) {
    Node *root = GetRoot(x); // original root
    MakeRoot(x);
    Access(y); // make x to root, tie with y
    Splay(x);
    x->lz += val;
    x->Push();
    MakeRoot(root); // Revert
    // edge update without edge vertex...
    Node *lca = GetLCA(x, y);
    Access(lca);
    Splay(lca);
    lca->Push();
    lca->Update(lca->now - val);
}
T VertexQuery(Node *x, Node *y) {
    Node *l = GetLCA(x, y);
    T ret = l->now;
    Access(x);
    Splay(l);
    if (l->r)

```

```

    ret = ret + l->r->sum;
    Access(y);
    Splay(l);
    if (l->r)
        ret = ret + l->r->sum;
    return ret;
}
Node *GetQueryResultNode(Node *u, Node *v) {
    if (!IsConnect(u, v))
        return 0;
    MakeRoot(u);
    Access(v);
    auto ret = v->l;
    while (ret->mx != ret->now) {
        if (ret->l && ret->mx == ret->l->mx)
            ret = ret->l;
        else
            ret = ret->r;
    }
    Access(ret);
    return ret;
} // code from justicehui

```

2 Graph & Flow

2.1 Hopcroft-Karp & König's

Usage: Dinic's variant. Maximum Matching = Minimum Vertex Cover = S - Maximum Independence Set

Time Complexity: $\mathcal{O}(\sqrt{V}E)$

```

while (true) {
    vector<int> level(sz, -1);
    queue<int> q;
    for (int x : l) {
        if (match[x] == -1) {
            level[x] = 0;
            q.push(x);
        }
    }
}

```

```

while (!q.empty()) {
    const int x = q.front();
    q.pop();
    for (int next : e[x]) {
        if (match[next] != -1 && level[match[next]] == -1) {
            level[match[next]] = level[x] + 1;
            q.push(match[next]);
        }
    }
}
if (level.empty() || *max_element(level.begin(), level.end()) == -1)
    break;
function<bool(int)> dfs = [&](int x) {
    for (int next : e[x]) {
        if (match[next] == -1 ||
            (level[match[next]] == level[x] + 1 &&
             dfs(match[next]))) {
            match[next] = x;
            match[x] = next;
            return true;
        }
    }
    return false;
};
int total = 0;
for (int x : l) if (level[x] == 0) total += dfs(x);
if (total == 0) break;
flow += total;
}
set<int> alt; // Konig
function<void(int, bool)> dfs = [&](int x, bool left) {
    if (alt.contains(x)) return;
    alt.insert(x);
    for (int next : e[x]) {
        if ((next != match[x]) && left) dfs(next, false);
        if ((next == match[x]) && !left) dfs(next, true);
    }
};
for (int x : l) if (match[x] == -1) dfs(x, true);

```

```

int test = 0;
for (int i : l) {
    if (alt.contains(i)) {
        auto &[y, x] = pos[i];
        s[y][x] = 'C';
    }
}
for (int i : r) {
    if (!alt.contains(i)) {
        auto &[y, x] = pos[i];
        s[y][x] = 'C';
    }
}

```

2.2 Dinic's

Time Complexity: $\mathcal{O}(V^2E)$, $\mathcal{O}(\min(V^{2/3}E, E^{3/2}))$ on unit capacity

```

while (true) {
    vector<int> level(dt, -1);
    queue<int> q;
    level[st] = 0;
    q.push(st);
    while (!q.empty()) {
        const int x = q.front();
        q.pop();
        for (int nid : eid[x]) {
            const auto &[_ , next, cap, flow] = e[nid];
            if (level[next] == -1 && cap - flow > 0) {
                level[next] = level[x] + 1;
                q.push(next);
            }
        }
    }
    if (level[dt] == -1) break;
    vector<int> vis(dt);
    function<int(int, int)> dfs = [&](int x, int total) {
        if (x == dt) return total;
        for (int &i = vis[x]; i < eid[x].size(); ++i) {
            auto &[_ , next, cap, flow] = e[eid[x][i]];
            if (level[next] == level[x] + 1 && cap - flow > 0) {

```

```

                const int res = dfs(next, min(total, cap - flow));
                if (res > 0) {
                    auto &[_next, _x, bcap, bflow] = e[eid[x][i] ^ 1];
                    assert(next == _next && x == _x);
                    flow += res;
                    bflow -= res;
                    return res;
                }
            }
        }
        return 0;
    };
    while (true) {
        const int res = dfs(st, 1e9 + 7);
        if (res == 0) break;
        ans += res;
    }
}

```

2.3 Strongly Connected Component

Time Complexity: $\mathcal{O}(N)$

```

int idx = 0, scnt = 0;
vector<int> scc(n, -1), vis(n, -1), st;
function<int(int)> dfs = [&](int x) {
    int ret = vis[x] = idx++;
    st.push_back(x);
    for (int next : e[x]) {
        if (vis[next] == -1)
            ret = min(ret, dfs(next));
        else if (scc[next] == -1)
            ret = min(ret, vis[next]);
    }
    if (ret == vis[x]) {
        while (!st.empty()) {
            const int t = st.back();
            st.pop_back();
            scc[t] = scnt;
            if (t == x)
                break;

```

```

    }
    scnt++;
}
return ret;
};

```

2.4 Biconnected Component

Time Complexity: $\mathcal{O}(N)$

```

int idx = 0;
vector<int> vis(n, -1);
vector<pii> st;
vector<vector<pii>> bcc;
vector<bool> cut(n); // articulation point
function<int (int, int)> dfs = [&] (int x, int p) {
    int ret = vis[x] = idx++;
    int child = 0;
    for (int next : e[x]) {
        if (next == p)
            continue;
        if (vis[next] < vis[x])
            st.emplace_back(x, next);
        if (vis[next] != -1)
            ret = min(ret, vis[next]);
        else {
            int res = dfs(next, x);
            ret = min(ret, res);
            child++;
            if (vis[x] ≤ res) {
                if (p != -1)
                    cut[x] = true;
                bcc.emplace_back();
                while (st.back() != pii{x, next}) {
                    bcc.back().push_back(st.back());
                    st.pop_back();
                }
                bcc.back().push_back(st.back());
                st.pop_back();
            } // vis[x] < res to find bridges
        }
    }
}

```

```

}
if (p == -1 && child > 1)
    cut[x] = true;
return ret;
};

```

2.5 Heavy-Light Decomposition

Usage: Query with the ETT number and it's root node

Time Complexity: $\mathcal{O}(N) - \mathcal{O}(\log N)$

```

vector<int> par(n), ett(n), rt(n), d(n), sz(n);
function<void (int)> dfs1 = [&] (int x) {
    sz[x] = 1;
    for (int &next : e[x]) {
        if (next == par[x]) continue;
        d[next] = d[x] + 1;
        par[next] = x;
        dfs1(next);
        sz[x] += sz[next];
        if (e[x][0] == par[x] || sz[e[x][0]] < sz[next])
            swap(e[x][0], next);
    }
};
int idx = 1;
function<void (int)> dfs2 = [&] (int x) {
    ett[x] = idx++;
    for (int next : e[x]) {
        if (next == par[x]) continue;
        rt[next] = next == e[x][0] ? rt[x] : next;
        dfs2(next);
    }
};

```

2.6 Centroid Decomposition

Usage: cent[x] is the parent in centroid tree

Time Complexity: $\mathcal{O}(N \log N)$

```

vector<int> sz(n);
vector<bool> fin(n);

```

```

function<int (int, int)> get_size = [&] (int x, int p) {
    sz[x] = 1;
    for (int next : e[x])
        if (!fin[next] && next != p) sz[x] += get_size(next, x);
    return sz[x];
};
function<int (int, int, int)> get_cent = [&] (int x, int p, int
all) {
    for (int next : e[x])
        if (!fin[next] && next != p && sz[next]*2 > all) return
get_cent(next, x, all);
    return x;
};
vector<int> cent(n, -1);
function<void (int, int)> get_cent_tree = [&] (int x, int p) {
    get_size(x, p);
    x = get_cent(x, p, sz[x]);
    fin[x] = true;
    cent[x] = p;
    function<void (int, int, int, bool)> dfs = [&] (int x, int p,
int d, bool test) {
        if (test) // update answer
        else // update state
        for (int next : e[x])
            if (!fin[next] && next != p) dfs(next, x, d, test);
    };
    for (int next : e[x]) {
        if (!fin[next]) {
            dfs(next, x, init, true);
            dfs(next, x, init+curr, false);
        }
    }
    for (int next : e[x])
        if (!fin[next] && next != p) get_cent_tree(next, x);
};
get_cent_tree(0, -1);

```

3 Geometry

3.1 Line intersection

Usage: Check the intersection of (x_1, x_2) and (y_1, y_2) . It requires an additional condition when they are parallel

Time Complexity: $\mathcal{O}(1)$

$ccw(x1, x2, y1) \neq ccw(x1, x1, y2) \ \&\& \ ccw(y1, y2, x1) \neq ccw(y1, y2, x2)$

3.2 Graham Scan

Time Complexity: $\mathcal{O}(N \log N)$

```

struct point {
    int x, y, p, q;
    point() { x = y = p = q = 0; }
    bool operator < (const point& other) {
        if (1LL * other.p * q != 1LL * p * other.q)
            return 1LL * other.p * q < 1LL * p * other.q;
        else if (y != other.y)
            return y < other.y;
        else
            return x < other.x;
    }
};
swap(points[0], *min_element(points.begin(), points.end()));
for (int i=1; i<points.size(); ++i) {
    points[i].p = points[i].x - points[0].x;
    points[i].q = points[i].y - points[0].y;
}
sort(points.begin()+1, points.end());
vector<int> hull;
for (int i=0; i<points.size(); ++i) {
    while (hull.size() ≥ 2 && ccw(points[hull[hull.size()-2]],
points[hull.back()], points[i]) < 1)
        hull.pop_back();
    hull.push_back(i);
}

```

3.3 Rotating Calipers

Usage: Get the maximum distance of the convex hull

Time Complexity: $\mathcal{O}(N)$

```
auto ccw4 = [&] (point& a1, point& a2, point& b1, point& b2) {
    return 1LL * (a2.x - a1.x) * (b2.y - b1.y) > 1LL * (a2.y - a1.y) * (b2.x - b1.x);
};
auto dist = [] (point& a, point& b) {
    return 1LL * (a.x - b.x) * (a.x - b.x) + 1LL * (a.y - b.y) * (a.y - b.y);
};
ll maxi = 0;
for (int i=0, j=1; i<hull.size(); i++) {
    maxi = max(maxi, dist(hull[i], hull[j]));
    if (j < hull.size()-1 && ccw4(hull[i], hull[i+1], hull[j], hull[j+1]))
        j++;
    else
        i++;
}
```

3.4 Bulldozer Trick

Usage: Traverse the entire sorting state of 2D points

Time Complexity: $\mathcal{O}(N^2 \log N)$

```
struct Line{
    ll i, j, dx, dy; // dx ≥ 0
    Line(int i, int j, const Point &pi, const Point &pj) {
        : i(i), j(j), dx(pj.x-pi.x), dy(pj.y-pi.y) {}
    bool operator < (const Line &l) const {
        return make_tuple(dy*l.dx, i, j) < make_tuple(l.dy*dx, l.i, l.j);
    }
    bool operator = (const Line &l) const {
        return dy * l.dx == l.dy * dx;
    }
};
void Solve(){
```

```
sort(A+1, A+N+1); iota(P+1, P+N+1, 1);
vector<Line> V; V.reserve(N*(N-1)/2);
for(int i=1; i≤N; i++) for(int j=i+1; j≤N; j++)
    V.emplace_back(i, j, A[i], A[j]);
sort(V.begin(), V.end());
for(int i=0, j=0; i<V.size(); i=j){
    while(j < V.size() && V[i] == V[j]) j++;
    for(int k=i; k<j; k++){
        int u = V[k].i, v = V[k].j; // point id, index -> Pos[id]
        swap(Pos[u], Pos[v]); swap(A[Pos[u]], A[Pos[v]]);
        if(Pos[u] > Pos[v]) swap(u, v);
        // @TODO
    }
}
} // code from justicehui
```

3.5 Point in Convex Polygon

Time Complexity: $\mathcal{O}(\log N)$

```
bool onsegment(pii a, pii b, pii c) {
    return ccw(a, b, c) == 0 && (a - c) * (b - c) ≤ 0;
}
bool pointinhull(pii* H, int n, pii p, bool strict = true) {
    int a = 1, b = n - 1, r = !strict;
    if (n < 3) return r && onsegment(H[0], H[n - 1], p);
    if (sign(ccw(H[0], H[a], H[b])) > 0) swap(a, b);
    if (sign(ccw(H[0], H[a], p)) ≥ r || sign(ccw(H[0], H[b], p) ≤ -r))
        return false;
    while (abs(a - b) > 1) {
        int c = (a + b) / 2;
        (sign(ccw(H[0], H[c], p)) > 0 ? b : a) = c;
    }
    return sign(ccw(H[a], H[b], p)) < r;
}
```

3.6 Line Hull Intersection

Time Complexity: $\mathcal{O}(\log N)$

```

/*
 * lineHull(line, poly) returns a pair describing the intersection
 * of a line with the polygon:
 * (-1, -1) if no collision,
 * (i, -1) if touching the corner $i$,
 * (i, i) if along side $(i, i+1)$,
 * (i, j) if crossing sides $(i, i+1)$ and $(j, j+1)$.
 * In the last case, if a corner $i$ is crossed, this is treated
 * as happening on side $(i, i+1)$.
 * The points are returned in the same order as the line hits the
 * polygon.
 * extrVertex returns the point of a hull with the max projection
 * onto a line.
 */
T cross(P p) const { return x*p.y - y*p.x; }
T cross(P a, P b) const { return (a-*this).cross(b-*this); }
P perp() const { return P(-y, x); } // rotates +90 degrees

#define cmp(i,j) sign(dir.perp().cross(poly[(i)%n]-poly[(j)%n]))
#define extr(i) cmp(i + 1, i) ≥ 0 && cmp(i, i - 1 + n) < 0
template <class P> int extrVertex(vector<P>& poly, P dir) {
    int n = sz(poly), lo = 0, hi = n;
    if (extr(0)) return 0;
    while (lo + 1 < hi) {
        int m = (lo + hi) / 2;
        if (extr(m)) return m;
        int ls = cmp(lo + 1, lo), ms = cmp(m + 1, m);
        (ls < ms || (ls == ms && ls == cmp(lo, m)) ? hi : lo) = m;
    }
    return lo;
}

#define cmpL(i) sign(a.cross(poly[i], b))
template <class P>
array<int, 2> lineHull(P a, P b, vector<P>& poly) {
    int endA = extrVertex(poly, (a - b).perp());
    int endB = extrVertex(poly, (b - a).perp());
    if (cmpL(endA) < 0 || cmpL(endB) > 0)
        return {-1, -1};
    array<int, 2> res;

```

```

for (int i = 0; i < 2; i++) {
    int lo = endB, hi = endA, n = sz(poly);
    while ((lo + 1) % n != hi) {
        int m = ((lo + hi + (lo < hi ? 0 : n)) / 2) % n;
        (cmpL(m) == cmpL(endB) ? lo : hi) = m;
    }
    res[i] = (lo + !cmpL(hi)) % n;
    swap(endA, endB);
}
if (res[0] == res[1]) return {res[0], -1};
if (!cmpL(res[0]) && !cmpL(res[1]))
    switch ((res[0] - res[1] + sz(poly) + 1) % sz(poly)) {
        case 0: return {res[0], res[0]};
        case 2: return {res[1], res[1]};
    }
return res;
}

```

4 Fast Fourier Transform

4.1 Fast Fourier Transform

Usage: FFT and multiply polynomials

Time Complexity: $\mathcal{O}(N \log N)$

```

#include <string>
#pragma GCC optimize("O3")
#pragma GCC target("avx,avx2,fma")
#include <bits/stdc++.h>
#include <immintrin.h>
#include <smintrin.h>
__m256d mult(__m256d a, __m256d b) {
    __m256d c = _mm256_movedup_pd(a);
    __m256d d = _mm256_shuffle_pd(a, a, 15);
    __m256d cb = _mm256_mul_pd(c, b);
    __m256d db = _mm256_mul_pd(d, b);
    __m256d e = _mm256_shuffle_pd(db, db, 5);
    __m256d r = _mm256_addsub_pd(cb, e);
    return r;
}

```

```

void fft(int n, __m128d a[], bool invert) {
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n >> 1;
        for (; j ≥ bit; bit >>= 1) j -= bit;
        j += bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len ≤ n; len <<= 1) {
        double ang = 2 * 3.14159265358979 / len * (invert ? -1 : 1);
        __m256d wlen;
        wlen[0] = cos(ang), wlen[1] = sin(ang);
        for (int i = 0; i < n; i += len) {
            __m256d w; w[0] = 1; w[1] = 0;
            for (int j = 0; j < len / 2; ++j) {
                w = _mm256_permute2f128_pd(w, w, 0);
                wlen = _mm256_insertf128_pd(wlen, a[i + j + len / 2], 1);
                w = mult(w, wlen);
                __m128d vw = _mm256_extractf128_pd(w, 1);
                __m128d u = a[i + j];
                a[i + j] = _mm_add_pd(u, vw);
                a[i + j + len / 2] = _mm_sub_pd(u, vw);
            }
        }
    }
    if (invert) {
        __m128d inv; inv[0] = inv[1] = 1.0 / n;
        for (int i = 0; i < n; ++i) a[i] = _mm_mul_pd(a[i], inv);
    }
}

vector<int64_t> multiply(vector<int64_t> &v, vector<int64_t> &w) {
    int n = 2;
    while (n < v.size() + w.size()) n <<= 1;
    __m128d *fv = new __m128d[n];
    for (int i = 0; i < n; ++i) fv[i][0] = fv[i][1] = 0;
    for (int i = 0; i < v.size(); ++i) fv[i][0] = v[i];
    for (int i = 0; i < w.size(); ++i) fv[i][1] = w[i];
    fft(n, fv, 0); // (a+bi) is stored in FFT
    for (int i = 0; i < n; i += 2) {
        __m256d a;
        a = _mm256_insertf128_pd(a, fv[i], 0);

```

```

        a = _mm256_insertf128_pd(a, fv[i + 1], 1);
        a = mult(a, a);
        fv[i] = _mm256_extractf128_pd(a, 0);
        fv[i + 1] = _mm256_extractf128_pd(a, 1);
    }
    fft(n, fv, 1);
    vector<int64_t> ret(n);
    for (int i = 0; i < n; ++i) ret[i] = (int64_t)round(fv[i][1] / 2);
    delete[] fv;
    return ret;
}

```

4.2 Number Theoretic Transform and Kitamasa

Usage: FFT with integer - to get better accuracy

Time Complexity: $\mathcal{O}(N \log N)$

```

// w is the root of mod e.g. 3/998244353 and 5/1012924417
void ntt(vector<ll> &f, const ll w, const ll mod) {
    const int n = f.size();
    if (n == 1)
        return;
    vector<ll> odd(n/2), even(n/2);
    for (int i=0; i<n; ++i)
        (i&1 ? odd : even)[i/2] = f[i];
    ntt(odd, w*w%mod, mod);
    ntt(even, w*w%mod, mod);
    ll x = 1;
    for (int i=0; i<n/2; ++i) {
        f[i] = (even[i] + x * odd[i] % mod) % mod;
        f[i+n/2] = (even[i] - x * odd[i] % mod + mod) % mod;
        x = x*w%mod;
    }
}

vector<int> mult(vector<int> f, vector<int> g) {
    int sz;
    for (sz = 1; sz < f.size() + g.size(); sz *= 2);
    vector<int> ret(sz);
    f.resize(sz), g.resize(sz);
    int w = modpow(W, (MOD - 1) / sz, MOD);

```

```

ntt(f, w), ntt(g, w);
for (int i = 0; i < sz; ++i)
    ret[i] = 1LL * f[i] * g[i] % MOD;
ntt(ret, modpow(w, MOD - 2, MOD));
const int szinv = modpow(sz, MOD - 2, MOD);
for (int i = 0; i < sz; ++i)
    ret[i] = 1LL * ret[i] * szinv % MOD;
while (!ret.empty() && ret.back() == 0)
    ret.pop_back();
return ret;
}

vector<int> inv(vector<int> f, const int DMOD) {
    vector<int> ret = {modpow(f[0], MOD - 2, MOD)};
    for (int i = 1; i < DMOD; i *= 2) {
        vector<int> tmp(f.begin(), f.begin() + min((int)f.size(), i * 2));
        tmp = mult(ret, tmp);
        tmp.resize(i * 2);
        for (int &x : tmp) x = (MOD - x) % MOD;
        tmp[0] = (tmp[0] + 2) % MOD;
        ret = mult(ret, tmp);
        ret.resize(i * 2);
    }
    ret.resize(DMOD);
    return ret;
}

vector<int> div(vector<int> a, vector<int> b) {
    if (a.size() < b.size()) return {};
    const int DMOD = a.size() - b.size() + 1;
    reverse(a.begin(), a.end());
    reverse(b.begin(), b.end());
    if (a.size() > DMOD) a.resize(DMOD);
    if (b.size() > DMOD) b.resize(DMOD);
    b = inv(b, DMOD);
    auto res = mult(a, b);
    res.resize(DMOD);
    reverse(res.begin(), res.end());
    while (!res.empty() && res.back() == 0) res.pop_back();
    return res;
}

```

```

vector<int> mod(vector<int> &a, vector<int> b) {
    auto tmp = mult(div(a, b), b);
    tmp.resize(a.size());
    for (int i = 0; i < a.size(); ++i)
        a[i] = (a[i] - tmp[i] + MOD) % MOD;
    while (!a.empty() && a.back() == 0) a.pop_back();
    return a;
}

vector<int> res = {1}, xn = {0, 1};
while (n) {
    if (n & 1) res = mod(mult(res, xn), c);
    n /= 2;
    xn = mod(mult(xn, xn), c);
}

```

4.3 Fast Walsh Hadamard Transform

Usage: XOR convolution

Time Complexity: $\mathcal{O}(N \log N)$

```

void fwht(vector<ll> &f) {
    const int n = f.size();
    if (n == 1)
        return;
    vector<ll> odd(n/2), even(n/2);
    for (int i=0; i<n; ++i)
        (i&1 ? odd : even)[i/2] = f[i];
    fwht(odd);
    fwht(even);
    for (int i=0; i<n/2; ++i) {
        f[i*2] = even[i] + odd[i];
        f[i*2+1] = even[i] - odd[i];
    }
}

```

5 String

5.1 Knuth-Moris-Pratt

Time Complexity: $\mathcal{O}(N)$

```
vector<int> fail(m);
for (int i=1, j=0; i<m; ++i) {
    while (j > 0 && p[i] != p[j]) j = fail[j-1];
    if (p[i] == p[j]) fail[i] = ++j;
}
vector<int> ans;
for (int i=0, j=0; i<n; ++i) {
    while (j > 0 && t[i] != p[j]) j = fail[j-1];
    if (t[i] == p[j]) {
        if (j == m-1) {
            ans.push_back(i-j);
            j = fail[j];
        } else j++;
    }
}
```

5.2 Rabin-Karp

Usage: The Rabin fingerprint for const-length hashing

Time Complexity: $\mathcal{O}(N)$

```
ull hash, p;
vector<ull> ht;
for (int i=0; i<=l-mid; ++i) {
    if (i == 0) {
        hash = s[0];
        p = 1;
        for (int j=1; j<mid; ++j) {
            hash = hash * pi + s[j];
            p = p * pi; // pi is the prime e.g. 13
        }
    } else
        hash = (hash - p * s[i-1]) * pi + s[i+mid-1];
    ht.push_back(hash);
}
```

5.3 Manacher

Usage: Longest radius of palindrome substring

Time Complexity: $\mathcal{O}(N)$

```
vector<int> man(m);
int r = 0, p = 0;
for (int i=0; i<m; ++i) {
    if (i <= r)
        man[i] = min(man[p*2 - i], r - i);
    while (i-man[i] > 0 && i+man[i] < m-1 && v[i-man[i]-1] ==
v[i+man[i]+1])
        man[i]++;
    if (r < i + man[i]) {
        r = i + man[i];
        p = i;
    }
}
```

5.4 Suffix Array and LCP Array

Time Complexity: $\mathcal{O}(N \log N) - \mathcal{O}(N)$

```
const int m = max(255, n)+1;
vector<int> sa(n), ord(n*2), nord(n*2);
for (int i=0; i<n; ++i) {
    sa[i] = i;
    ord[i] = s[i];
}
for (int d=1; d<n; d*=2) {
    auto cmp = [&](int i, int j) {
        if (ord[i] == ord[j])
            return ord[i+d] < ord[j+d];
        return ord[i] < ord[j];
    };
    vector<int> cnt(m), tmp(n);
    for (int i=0; i<n; ++i)
        cnt[ord[i+d]]++;
    for (int i=0; i+1<m; ++i)
        cnt[i+1] += cnt[i];
    for (int i=n-1; i>=0; --i)
        tmp[--cnt[ord[i+d]]] = i;
    fill(cnt.begin(), cnt.end(), 0);
    for (int i=0; i<n; ++i)
        cnt[ord[i]]++;
    for (int i=0; i+1<m; ++i)
```

```

    cnt[i+1] += cnt[i];
    for (int i=n-1; i>=0; --i)
        sa[--cnt[ord[tmp[i]]]] = tmp[i];
    nord[sa[0]] = 1;
    for (int i=1; i<n; ++i)
        nord[sa[i]] = nord[sa[i-1]] + cmp(sa[i-1], sa[i]);
    swap(ord, nord);
}
vector<int> inv(n), lcp(n);
for (int i=0; i<n; ++i)
    inv[sa[i]] = i;
for (int i=0, k=0; i<n; ++i) {
    if (inv[i] == 0)
        continue;
    for (int j=sa[inv[i]-1]; max(i,j)+k<n&&sa[i+k]==s[j+k]; ++k);
    lcp[inv[i]] = k ? k-- : 0;
}

```

5.5 Suffix Automaton

Usage: Suffix link corresponds to suffix tree of $\text{rev}(S)$

Time Complexity: $\mathcal{O}(N) - \mathcal{O}(N)$ using hashmap or $\mathcal{O}(1)$ size array

```

struct suffix_automaton {
    struct node {
        int len, slink;
        map<int, int> go;
    };
    int last = 0;
    vector<node> sa = {{0, -1}};
    void insert(int x) {
        sa.emplace_back(sa[last].len + 1, 0);
        int p = last;
        last = sa.size() - 1;
        while (p != -1 && !sa[p].go.contains(x))
            sa[p].go[x] = last, p = sa[p].slink;
        if (p != -1) {
            const int t = sa[p].go[x];
            if (sa[p].len + 1 < sa[t].len) {
                const int q = sa.size();
                sa.push_back(sa[t]);
            }
        }
    }
}

```

```

        sa[q].len = sa[p].len + 1;
        sa[t].slink = q;
        while (p != -1 && sa[p].go[x] == t)
            sa[p].go[x] = q, p = sa[p].slink;
        sa[last].slink = q;
    } else
        sa[last].slink = t;
    }
}
};

```

5.6 Aho-Corasick

Time Complexity: $\mathcal{O}(N + \sum M)$

```

struct trie {
    array<trie*, 3> go;
    trie* fail;
    int output, idx;
    trie() {
        fill(go.begin(), go.end(), nullptr);
        fail = nullptr;
        output = idx = 0;
    }
    ~trie() {
        for (auto &x : go)
            delete x;
    }
    void insert(const string &input, int i) {
        if (i == input.size())
            output++;
        else {
            const int x = input[i] - 'A';
            if (!go[x])
                go[x] = new trie();
            go[x]->insert(input, i+1);
        }
    }
};
queue<trie*> q; // make fail links; requires root->insert before

```

```

root->fail = root;
q.push(root);
while (!q.empty()) {
    trie *curr = q.front();
    q.pop();
    for (int i=0; i<26; ++i) {
        trie *next = curr->go[i];
        if (!next)
            continue;
        if (curr == root)
            next->fail = root;
        else {
            trie *dest = curr->fail;
            while (dest != root && !dest->go[i])
                dest = dest->fail;
            if (dest->go[i])
                dest = dest->go[i];
            next->fail = dest;
        }
        if (next->fail->output)
            next->output = true;
        q.push(next);
    }
}
trie *curr = root; // start query
bool found = false;
for (char c : s) {
    c -= 'a';
    while (curr != root && !curr->go[c])
        curr = curr->fail;
    if (curr->go[c])
        curr = curr->go[c];
    if (curr->output) {
        found = true;
        break;
    }
}
}

```

6 DP Optimization

6.1 Convex Hull Trick w/ Stack

Usage: $dp[i] = \min(dp[j] + b[j] * a[i]), b[j] \geq b[j+1]$

Time Complexity: $\mathcal{O}(N \log N) - \mathcal{O}(N)$ where $a[i] \leq a[i+1]$

```

struct lin {
    ll a, b; double s;
    ll f(ll x) { return a*x + b; }
};
inline double cross(const lin &x, const lin &y) {
    return 1.0 * (x.b - y.b) / (y.a - x.a);
}
vector<ll> dp(n);
vector<lin> st;
for (int i=1; i<n; ++i) {
    lin curr = { b[i-1], dp[i-1], 0 };
    while (!st.empty()) {
        curr.s = cross(st.back(), curr);
        if (st.back().s < curr.s) break;
        st.pop_back();
    }
    st.push_back(curr);
    int x = -1;
    for (int y = st.size(); y > 0; y /= 2)
        while (x+y < st.size() && st[x+y].s < a[i]) x += y;
    dp[i] = s[x].f(a[i]);
}
while (x+1 < st.size() && st[x+1].s < a[i]) ++x; // O(N) case

```

6.2 Convex Hull Trick w/ Li-Chao Tree

Usage: `update(l, r, 0, { a, b })`

Time Complexity: $\mathcal{O}(N \log N)$

```

static constexpr ll INF = 2e18;
struct lin {
    ll a, b;
    ll f(ll x) { return a*x + b; }
};

```

```

struct Lichao {
    struct node {
        int l, r; lin line;
    };
    vector<node> tree;
    void init() { tree.push_back({-1, -1, { 0, -INF }}); }
    void update(ll s, ll e, int n, const lin &line) {
        lin hi = tree[n].line;
        lin lo = line;
        if (hi.f(s) < lo.f(s)) swap(lo, hi);
        if (hi.f(e) ≥ lo.f(e)) {
            tree[n].line = hi;
            return;
        }
        const ll m = s + e >> 1;
        if (hi.f(m) > lo.f(m)) {
            tree[n].line = hi;
            if (tree[n].r == -1) {
                tree[n].r = tree.size();
                tree.push_back({-1, -1, { 0, -INF }});
            }
            update(m+1, e, tree[n].r, lo);
        } else {
            tree[n].line = lo;
            if (tree[n].l == -1) {
                tree[n].l = tree.size();
                tree.push_back({-1, -1, { 0, -INF }});
            }
            update(s, m, tree[n].l, hi);
        }
    }
    ll query(ll s, ll e, int n, ll x) {
        if (n == -1) return -INF;
        const ll m = s + e >> 1;
        if (x ≤ m) return max(tree[n].line.f(x), query(s, m, tree[n].l, x));
        else return max(tree[n].line.f(x), query(m+1, e, tree[n].r, x));
    }
};

```

6.3 Divide and Conquer Optimization

Usage: $dp[t][i] = \min(dp[t-1][j] + c[j][i])$, c is Monge

Time Complexity: $\mathcal{O}(KN \log N)$

```

vector<vector<ll>> dp(n, vector<ll>(t));
function<void (int, int, int, int, int)> dnc = [&] (int l, int r,
int s, int e, int u) {
    if (l > r)
        return;
    const int mid = (l + r) / 2;
    int opt;
    for (int i=s; i≤min(e, mid); ++i) {
        ll x = sum[i][mid] + C;
        if (i && u)
            x += dp[i-1][u-1];
        if (x ≥ dp[mid][u]) {
            dp[mid][u] = x;
            opt = i;
        }
    }
    dnc(l, mid-1, s, opt, u);
    dnc(mid+1, r, opt, e, u);
};
for (int i=0; i<t; ++i)
    dnc(0, n-1, 0, n-1, i);

```

6.4 Monotone Queue Optimization

Usage: $dp[i] = \min(dp[j] + c[j][i])$, c is Monge, find cross

Time Complexity: $\mathcal{O}(N \log N)$

```

auto cross = [&](ll p, ll q) {
    ll lo = min(p, q) - 1, hi = n + 1;
    while (lo + 1 < hi) {
        const ll mid = (lo + hi) / 2;
        if (f(p, mid) < f(q, mid)) lo = mid;
        else hi = mid;
    }
    return hi;
};

```

```

deque<pll> st;
for (int i = 1; i ≤ n; ++i) {
    pll curr{i - 1, 0};
    while (!st.empty() &&
           (curr.second = cross(st.back().first, i - 1)) ≤
            st.back().second)
        st.pop_back();
    st.push_back(curr);
    while (st.size() > 1 && st[1].second ≤ i) st.pop_front();
    dp[i] = f(st[0].first, i);
}

```

6.5 Aliens Trick

Usage: $dp[t][i] = \min(dp[t-1][j] + c[j+1][i])$, c is Monge, find lambda w/ half bs

Time Complexity: $\mathcal{O}(N \log N)$

```

ll lo = 0, hi = 1e15;
while (lo + 1 < hi) {
    const ll mid = (lo + hi) / 2;
    auto [dp, cnt] = dec(mid); // the best DP[N][K] and its K
    value
    if (cnt < k) hi = mid;
    else lo = mid;
}
cout << (dec(lo).first - lo * k) / 2;

```

6.6 Knuth Optimization

Usage: $dp[i] = \min(dp[i][k] + dp[k][j]) + c[i][j]$, Monge, Monotonic

Time Complexity: $\mathcal{O}(N^2)$

```

vector<vector<int>> dp(n, vector<int>(n)), opt(n, vector<int>(n));
for (int i=0; i<n; ++i)
    opt[i][i] = i;
for (int j=1; j<n; ++j) {
    for (int s=0; s<n-j; ++s) {
        int e = s+j;
        dp[s][e] = 1e9+7;

```

```

        for (int o=opt[s][e-1]; o<min(opt[s+1][e]+1, e); ++o) {
            if (dp[s][e] > dp[s][o] + dp[o+1][e]) {
                dp[s][e] = dp[s][o] + dp[o+1][e];
                opt[s][e] = o;
            }
        }
        dp[s][e] += sum[e+1] - sum[s];
    }
}

```

6.7 Sum Over Subsets

Usage: $dp[mask] = \text{sum}(A[i])$, i is in mask

Time Complexity: $\mathcal{O}(N2^N)$

```

for (int i=0; i<(1<<n); i++)
    f[i] = a[i];
for (int j=0; j<n; j++)
    for (int i=0; i<(1<<n); i++)
        if (i & (1<<j)) f[i] += f[i ^ (1<<j)];

```

7 Number Theory

7.1 Modular Operator

Usage: For Fermat's little theorem and Pollard rho

Time Complexity: $\mathcal{O}(\log N)$

```

using ull = unsigned long long;
ull modmul(ull a, ull b, ull n) { return ((unsigned __int128)a *
b) % n; }
ull modmul(ull a, ull b, ull n) { // if __int128 isn't available
    if (b == 0) return 0;
    if (b == 1) return a;
    ull t = modmul(a, b/2, n);
    t = (t+t)%n;
    if (b % 2) t = (t+a)%n;
    return t;
}
ull modpow(ull a, ull d, ull n) {

```

```

    if (d == 0) return 1;
    ull r = modpow(a, d/2, n);
    r = modmul(r, r, n);
    if (d % 2) r = modmul(r, a, n);
    return r;
}
ull gcd(ull a, ull b) { return b ? gcd(b, a%b) : a; }

```

7.2 Modular Inverse in $\mathcal{O}(N)$

Usage: Get inverse of factorial

Time Complexity: $\mathcal{O}(N) - \mathcal{O}(1)$

```

const int mod = 1e9+7;
vector<int> fact(n+1), inv(n+1), factinv(n+1);
fact[0] = fact[1] = inv[1] = factinv[0] = factinv[1] = 1;
for (int i=2; i≤n; ++i) {
    fact[i] = 1LL * fact[i-1] * i % mod;
    inv[i] = mod - 1LL * mod/i * inv[mod%i] % mod;
    factinv[i] = 1LL * factinv[i-1] * inv[i] % mod;
}

```

7.3 Extended Euclidean

Usage: get a and b as arguments and return the solution (x, y) of equation $ax + by = \gcd(a, b)$.

Time Complexity: $\mathcal{O}(\log a + \log b)$

```

pair<ll, ll> extGCD(ll a, ll b){
    if (b != 0) {
        auto tmp = extGCD(b, a % b);
        return {tmp.second, tmp.first - (a / b) * tmp.second};
    } else return {1ll, 0ll};
}

```

7.4 Floor Sum

Usage: sum of $\lfloor (ax + b)/c \rfloor$ where $x \in [0, n]$

Time Complexity: $\mathcal{O}(\log N)$

```

ll floor_sum(ll a, ll b, ll c, ll n) {
    ll ans = 0;
    if (a < 0) {
        ans -= (n * (n + 1) / 2) * ((a % c + c - a) / c);
        a = a % c + c;
    }
    if (b < 0) {
        ans -= (n + 1) * ((b % c + c - b) / c);
        b = b % c + c;
    }
    if (a == 0) return ans + b / c * (n + 1);
    if (a ≥ c or b ≥ c)
        return ans + (n * (n + 1) / 2) * (a / c) + (n + 1) * (b / c) +
            floor_sum(a % c, b % c, c, n);
    ll m = (a * n + b) / c;
    return ans + m * n - floor_sum(c, c - b - 1, a, m - 1);
}

```

7.5 Miller-Rabin

Usage: Fast prime test for big integers

Time Complexity: $\mathcal{O}(k \log N)$

```

bool is_prime(ull n) {
    const ull as[7] = {2, 325, 9375, 28178, 450775, 9780504,
        1795265022};
    // const ull as[12] = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
    37}; // easier to remember
    auto miller_rabin = [] (ull n, ull a) {
        ull d = n-1, temp;
        while (d % 2 == 0) {
            d /= 2;
            temp = modpow(a, d, n);
            if (temp == n-1)
                return true;
        }
        return temp == 1;
    };
    for (ull a : as) {
        if (a ≥ n)
            break;
    }
}

```

```

        if (!miller_rabin(n, a))
            return false;
    }
    return true;
}

```

7.6 Chinese Remainder Theorem

Usage: Solution for the system of linear congruence

Time Complexity: $\mathcal{O}(\log N)$

```

w1 = modpow(mod2, mod1-2, mod1);
w2 = modpow(mod1, mod2-2, mod2);
ll ans = ((__int128)mod2 * w1 * f1[i] + (__int128)mod1 * w2 *
f2[i]) % (mod1*mod2);

```

7.7 Pollard Rho

Usage: Factoring large numbers fast

Time Complexity: $\mathcal{O}(N^{1/4})$

```

void pollard_rho(ull n, vector<ull> &factors) {
    if (n == 1)
        return;
    if (n % 2 == 0) {
        factors.push_back(2);
        pollard_rho(n/2, factors);
        return;
    }
    if (is_prime(n)) {
        factors.push_back(n);
        return;
    }
    ull x, y, c = 1, g = 1;
    auto f = [&] (ull x) { return (modmul(x, x, n) + c) % n; };
    y = x = 2;
    while (g == 1 || g == n) {
        if (g == n) {
            c = rand() % 123;
            y = x = rand() % (n-2) + 2;
        }

```

```

        x = f(x);
        y = f(f(y));
        g = gcd(n, y>x ? y-x : x-y);
    }
    pollard_rho(g, factors);
    pollard_rho(n / g, factors);
}

```

8 ETC

8.1 Gaussian Elimination on $\mathbb{Z}_2^n$

Time Complexity: $\mathcal{O}(Nd^2/64)$

```

struct basis {
    const static int n = 30; // log2(1e9)
    array<int, n> data{};
    void insert(int x) {
        for (int i=0; i<n; ++i)
            if (data[i] && (x >> (n-1-i) & 1)) x ^= data[i];
        int y;
        for (y=0; y<n; ++y)
            if (!data[y] && (x >> (n-1-y) & 1)) break;
        if (y < n) {
            for (int i=0; i<n; ++i)
                if (data[i] >> (n-1-y) & 1) data[i] ^= x;
            data[y] = x;
        }
    }
    basis operator+(const basis &other) {
        basis ret{};
        for (int x : data) ret.insert(x);
        for (int x : other.data) ret.insert(x);
        return ret;
    }
};

```

8.2 Gaussian Elimination on $\mathbb{Z}_p^n$

Usage: Kirchhoff's, LGV, etc.

Time Complexity: $\mathcal{O}(N^3)$

```
int det(vector<vector<int>> a, const int mod) {
    const int n = a.size();
    assert(a[0].size() == n);
    int ret = 1;
    for (int j = 0; j < n; ++j) {
        int p = j;
        while (p < n && a[p][j] == 0) p++;
        if (p == n) return 0;
        if (p > j) {
            ret = mod - ret;
            swap(a[j], a[p]);
        }
        ret = 1LL * ret * a[j][j] % mod;
        const int inv = modpow(a[j][j], mod - 2, mod);
        for (int k = j; k < n; ++k)
            a[j][k] = 1LL * a[j][k] * inv % mod;
        for (int i = j + 1; i < n; ++i) {
            for (int k = n - 1; k >= 0; --k)
                a[i][k] = (a[i][k] - 1LL * a[i][j] * a[j][k] % mod + mod)
                    % mod;
        }
    }
    return ret;
}
```

8.3 Useful Stuff

- Catalan Number
 $1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, 208012, 742900$
 $C_n = \text{binomial}(n * 2, n) / (n + 1);$
 - 길이가 $2n$ 인 올바른 괄호 수식의 수
 - $n + 1$ 개의 리프를 가진 풀 바이너리 트리의 수
 - $n + 2$ 각형을 n 개의 삼각형으로 나누는 방법의 수
- Burnside's Lemma
 경우의 수를 세는데, 특정 transform operation(회전, 반사, ..) 해서 같은 경우들은 하나로 친다. 전체 경우의 수는? 각 operation마다 이 operation을 했을 때 변하지 않는 경우의 수를 센다 (단, “아무것도 하지 않는다” 라는 operation도

있어야 함!) 전체 경우의 수를 더한 후, operation의 수로 나눈다. (답이 맞다면 항상 나누어 떨어져야 한다)

- 알고리즘 게임
 - Nim Game의 해법: 각 더미의 돌의 개수를 모두 XOR했을 때 0 이 아니면 첫번째, 0 이면 두번째 플레이어가 승리.
 - Grundy Number: 어떤 상황의 Grundy Number는, 가능한 다음 상황들의 Grundy Number를 모두 모은 다음, 그 집합에 포함 되지 않는 가장 작은 수가 현재 state의 Grundy Number가 된다. 만약 다음 state가 독립된 여러개의 state들로 나뉘는 경우, 각각의 state의 Grundy Number의 XOR 합을 생각한다.
 - Subtraction Game: 한 번에 k 개까지의 돌만 가져갈 수 있는 경우, 각 더미의 돌의 개수를 $k + 1$ 로 나눈 나머지를 XOR 합하여 판단한다.
 - Index-k Nim: 한 번에 최대 k 개의 더미를 골라 각각의 더미에서 아무렇게나 돌을 제거할 수 있을 때, 각 binary digit에 대하여 합을 $k + 1$ 로 나눈 나머지를 계산한다. 만약 이 나머지가 모든 digit에 대하여 0이라면 두번째, 하나라도 0이 아니면 첫번째 플레이어가 승리.
- Pick's Theorem
 격자점으로 구성된 simple polygon이 주어짐. I 는 polygon 내부의 격자점 수, B 는 polygon 선분 위 격자점 수, A 는 polygon의 넓이라고 할 때, 다음과 같은 식이 성립한다. $A = I + B/2 - 1$
- 가장 가까운 두 점: 분할정복으로 가까운 6개의 점만 확인
- 홀의 결혼 정리: 이분그래프(L-R)에서, 모든 L을 매칭하는 필요충분 조건 = L에서 임의의 부분집합 S를 골랐을 때, 반드시 $(S \text{의 크기}) \leq (S \text{와 연결되어있는 모든 R의 크기})$ 이다.
- 소수: 10 007, 10 009, 10 111, 31 567, 70 001, 1 000 003, 1 000 033, 4 000 037, 99 999 989, 999 999 937, 1 000 000 007, 1 000 000 009, 9 999 999 967, 99 999 999 977
- 소수 개수: (1e5 이하: 9592), (1e7 이하: 664 579), (1e9 이하: 50 847 534)
- 10^{15} 이하의 정수 범위의 나눗셈 한번은 오차가 없다.
- N 의 약수의 개수 = $\mathcal{O}(N^{1/3})$, N 의 약수의 합 = $\mathcal{O}(N \log \log N)$
- $\phi(mn) = \phi(m)\phi(n)$, $\phi(pr^n) = pr^n - pr^{n-1}$, $a^{\phi(n)} \equiv 1 \pmod{n}$ if coprime
- Euler characteristic: $v - e + f$ (면, 외부 포함) = $1 + c$ (컴포넌트)
- Euler's phi $\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$

- Lucas' Theorem $\binom{m}{n} = \prod \binom{m_i}{n_i} \pmod{p}$ m_i, n_i 는 p^i 의 계수
- 스케줄링에서 데드라인이 빠른 걸 쓰는게 이득. 늦은 스케줄이 안들어갈 때 가장 시간 소모가 큰 스케줄 1개를 제거하면 이득.

8.4 Template

```
// precision
cout.precision(16);
cout << fixed;
// gcc bit operator
__builtin_popcount(bits); // popcountll for ll
__builtin_clz(bits);      // left
__builtin_ctz(bits);      // right
// random number generator
random_device rd;
mt19937 mt(rd()); // or use chrono
uniform_int_distribution<> half(0, 1);
cout << half(mt);
// 128MB = int * 33,554,432
struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        // http://xorshift.di.unimi.it/splitmix64.c
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM =
            chrono::steady_clock::now().time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <typename K, typename V, typename Comp = less<K>>
using ordered_map =
    tree<K, V, Comp, rb_tree_tag,
    tree_order_statistics_node_update>;
```

```
template <typename K, typename Comp = less<K>> // less_equal (MS)
using ordered_set = ordered_map<K, null_type, Comp>;
gp_hash_table<key, int, custom_hash> table;
regex re("^first.[0-9a-z]?*+{n}-{n,m}");
regex_match(s, re)
// debug macros
template <class T, class U>
ostream &operator<<(ostream &out, const pair<T, U> &v) {
    out << "(" << v.first << ', ' << v.second << ")";
    return out;
}
template <class... Ts>
ostream &operator<<(ostream &out, const tuple<Ts...> &v) {
    out << "(";
    [&<size_t... Is>(index_sequence<Is...>) {
        ((out << (Is == 0 ? "" : ",") << get<Is>(v)), ...);
    }(index_sequence_for<Ts...>{});
    out << ")";
    return out;
}
template <ranges::range T>
requires(!is_convertible_v<T, std::string>)
ostream &operator<<(ostream &out, const T &v) {
    out << '[';
    bool first = true;
    for (const auto &x : v) {
        if (!first) out << ", ";
        out << x;
        first = false;
    }
    out << ']';
    return out;
}
#ifdef ONLINE_JUDGE
#define debug(x) cout << "[Debug] " << #x << " = " << x << '\n'
#define dout cout
#else
#define debug(x) void(0)
#define dout if (false) cout
#endif
```

```
// CLion CMakeLists.txt
cmake_minimum_required(VERSION 3.30)
project(ps)
set(CMAKE_CXX_STANDARD 20)
include_directories(.)
MATH(EXPR stack_size "1024 * 1024 * 1024")
set(CMAKE_CXX_FLAGS "${CMAKE_CXX_FLAGS} -Wextra -Wall
-fsanitize=address -fsanitize=undefined")
set(CMAKE_EXE_LINKER_FLAGS "-Wl,--stack,${stack_size}")
add_definitions(-DLOCAL)
add_executable(ps main.cpp)
```

8.5 자주 쓰이는 문제 접근법

- 비슷한 문제를 풀어본 적이 있던가?
- 단순한 방법에서 시작할 수 있을까? (brute force)
- 내가 문제를 푸는 과정을 수식화할 수 있을까? (예제를 직접 해결해보면서)
- 문제를 단순화할 수 있을까?
- 그림으로 그려볼 수 있을까?
- 수식으로 표현할 수 있을까?
- 문제를 분해할 수 있을까?
- 뒤에서부터 생각해서 문제를 풀 수 있을까?
- 순서를 강제할 수 있을까?
- 특정 형태의 답만을 고려할 수 있을까? (정규화)
- 특수 조건을 꼭 활용
- 여사건으로 생각하기
- 게임이론 - 거울 전략 혹은 mex DP 연계
- 겁먹지 말고 경우 나누어 생각
- 해법에서 역순으로 가능한가?
- 딱 맞는 시간복잡도에 집착하지 말자

- 문제에 의미있는 작은 상수 이용
- 스몰투라지, 트라이, 해싱, 루트질 같은 트릭 생각
- 너무 추상화하기보단 풀려야 하는 방식으로 생각하기

8.6 DP 최적화 접근

- $C[i, j] = A[i] * B[j]$ 이고 A, B가 단조증가, 단조감소이면 Monge
- l..r의 값들의 sum 이나 min은 Monge
- 식 정리해서 일차(CHT) 혹은 비슷한(MQ) 함수를 발견, 구현 힘들면 Li-Chao
- $a \leq b \leq c \leq d$ 에서 $A[a, c] + A[b, d] \leq A[a, d] + A[b, c]$
- Monge 성질을 보이기 어려우면 N^2 나이트 짜서 opt의 단조성을 확인하고 적맞
- 식이 간단하거나 변수가 독립적이면 DP 테이블을 세그 위에 올려서 해결
- 침착하게 점화식부터 세우고 Monge인지 판별
- Monge에 집착하지 말고 단조성이나 볼록성만 보여도 됨

8.7 Fast I/O

```
#pragma GCC optimize("O3")
#pragma GCC optimize("Ofast")
#pragma GCC optimize("unroll-loops")

inline int readChar();
template<class T = int> inline T readInt();
template<class T> inline void writeInt(T x, char end = 0);
inline void writeChar(int x);
inline void writeWord(const char *s);
static const int buf_size = 1 << 18;
inline int getChar(){
    #ifndef LOCAL
        static char buf[buf_size];
        static int len = 0, pos = 0;
        if(pos == len) pos = 0, len = fread(buf, 1, buf_size, stdin);
        if(pos == len) return -1;
        return buf[pos++];
    #endif
}
```

```

}
inline int readChar(){
    #ifndef LOCAL
    int c = getChar();
    while(c ≤ 32) c = getChar();
    return c;
    #else
    char c; cin >> c; return c;
    #endif
}

template <class T>
inline T readInt(){
    #ifndef LOCAL
    int s = 1, c = readChar();
    T x = 0;
    if(c == '-') s = -1, c = getChar();
    while('0' ≤ c && c ≤ '9') x = x * 10 + c - '0', c =
    getChar();
    return s == 1 ? x : -x;
    #else
    T x; cin >> x; return x;
    #endif
}

static int write_pos = 0;
static char write_buf[buf_size];
inline void writeChar(int x){
    if(write_pos == buf_size) fwrite(write_buf, 1, buf_size,
    stdout), write_pos = 0;
    write_buf[write_pos++] = x;
}

template <class T>
inline void writeInt(T x, char end){
    if(x < 0) writeChar('-'), x = -x;
    char s[24]; int n = 0;
    while(x || !n) s[n++] = '0' + x % 10, x /= 10;
    while(n--) writeChar(s[n]);
    if(end) writeChar(end);
}

inline void writeWord(const char *s){
    while(*s) writeChar(*s++);
}

```

```

}
struct Flusher{
    ~Flusher(){ if(write_pos) fwrite(write_buf, 1, write_pos,
    stdout), write_pos = 0; }
}flusher;

```

8.8 Bitset Add Sub

```

#define private public
#include <bitset>
#undef private
#include <x86intrin.h>

template <size_t _Nw>
void _M_do_sub(_Base_bitset<_Nw> &A, const _Base_bitset<_Nw> &B) {
    for (int i = 0, c = 0; i < _Nw; i++)
        c = _subborrow_u64(c, A._M_w[i], B._M_w[i],
        (unsigned long long *)&A._M_w[i]);
}

template <size_t _Nb>
inline bitset<_Nb> operator-(const bitset<_Nb> &A, const
bitset<_Nb> &B) {
    bitset<_Nb> C(A);
    return C -= B;
}

template <size_t _Nw>
void _M_do_add(_Base_bitset<_Nw> &A, const _Base_bitset<_Nw> &B) {
    for (int i = 0, c = 0; i < _Nw; i++)
        c = _addcarry_u64(c, A._M_w[i], B._M_w[i],
        (unsigned long long *)&A._M_w[i]);
}

```